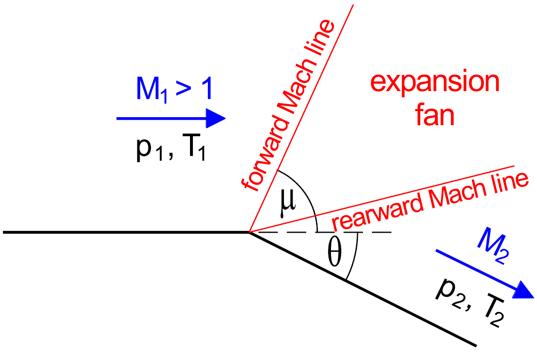
Prandtl-Meyer Expansion



Prandtl Meyer theory describes the isentropic expansion of a supersonic flow around an external corner.

The flow is assumed to turn continuously to follow the boundary through a series of infinitessimal Mach waves ($\sin\mu = 1/M$), maintaining constant total temperature, pressure and wave-tangential velocity component. The angle turned θ is then related to upstream and downstream Mach numbers by:

$$\theta = \nu(M_2) - \nu(M_1) \tag{1}$$

where the Prandtl-Meyer function ν is

$$\nu(M) = \alpha \tan^{-1}\left(\frac{\sqrt{M^2-1}}{\alpha}\right) - \tan^{-1}\sqrt{M^2-1} \tag{2}$$

where

$$\alpha = \sqrt{\frac{\gamma+1}{\gamma-1}} \tag{3}$$

and γ is the ratio of specific heat capacities. $\nu(M)$ is monotonic increasing and $\nu(1) = 0$.

Knowledge of upstream Mach number M_1 and angle turned θ gives, in sequence,

ν_1	(from (2))
ν_2	(from (1))
M_2	(inverting (1)

Total temperature T_0 and total pressure P_0 are related to local temperature and pressure by

$$\frac{T_0}{T} = 1 + \tfrac{1}{2}(\gamma-1)M^2, \qquad \frac{P_0}{P} = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}} \tag{4}$$

As T_0 and P_0 are constant in an isentropic expansion, the ratios of quantities either side of the expansion are given by

$$\frac{T_2}{T_1} = \frac{T_2/T_0}{T_1/T_0} = \frac{1+\tfrac{1}{2}(\gamma-1)M_1^2}{1+\tfrac{1}{2}(\gamma-1)M_2^2} \tag{5}$$

$$\frac{P_2}{P_1} = \frac{P_2/P_0}{P_1/P_0} = \left(\frac{T_2}{T_1}\right)^{\frac{\gamma}{\gamma-1}} \tag{6}$$

Nomenclature

- p = pressure;
- T = absolute temperature;
- c_p, c_v = specific heat capacities at constant pressure and volume, respectively;
- $\gamma \equiv c_p/c_v$ = ratio of specific heat capacities;
- M = Mach number.

Subscripts 1 and 2 denote upstream and downstream conditions, respectively.
Subscript 0 denotes total (or stagnation) conditions.